AMERICAN TERAWATT

Liquidated Damages

Availability & performance guarantees for HVDC systems

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Revision history

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0.1	August 4, 2026	Eugen Starschich	Initial docuemnt

Contents

1 Executive summary

This report examines **liquidated damages (LD) tied to availability and performance guarantees** for HVDC transmission systems, and how they should be framed for American Terawatt's projects supplying large industrial and data-center loads.

For an HVDC scheme, the commercially relevant guarantees are *operational*: the fraction of energy the link can transmit over a reporting period. The industry standard for measuring this is CIGRE Technical Brochure 956 (2025), which defines forced and scheduled energy unavailability (FEU, SEU) and the forced outage rate (FOR). When the measured value exceeds the contractually guaranteed threshold, liquidated damages become payable.

Key points

- **Guaranteed values (realistic, OEM-backable):** FEU $\approx$ 0.5 %, SEU $\approx$ 1 %, and a bipolar forced-outage rate of order 1 per 10 years. Very aggressive figures (e.g. 1 bipolar outages per 20 years) are *commercially driven, not technically justified*, and no OEM will guarantee them with high penalty costs.
- **LD market ranges (~1 GW project):** \$0.1–1.2 M per additional forced outage; \$75–600 k per increment of forced energy unavailability; overall LD cap of 3–5 % of contract value in the last years. Prior to that, LD cap of up to 20 % were very common.
- **Availability gap:** a single HVDC link cannot, on its own, meet a data-center Tier III target ($\approx$99.982 %). Meeting it requires redundancy (dual full-capacity circuits), on-site back-up generation, or a negotiated availability target — all of which shape the LD position.
- **LD design principles:** LDs should be expressed as a percentage of OEM supply value, set high enough to *incentivise the OEM to fix problems*, and understood as a behavioural lever rather than full recovery of loss.

Recommendation. Base the LD position on a defensible economic argument — the unavailability charges American Terawatt faces under its power purchase agreement (PPA) and the cost of the battery capacity required to compensate for forced-unavailability events — and argue for an LD cap linked to that overall PPA liability rather than defaulting to the standard 3–5 %.

2 Introduction and scope

2.1 Background

American Terawatt's HVDC schemes are being developed to supply large industrial and data-center loads. For such customers the value of the connection depends not only on its rating but on how reliably it can deliver energy over time. Availability and performance therefore become contractual guarantees, and **liquidated damages** are the mechanism by which a shortfall against those guarantees is priced.

2.2 Objective

This report sets out:

(i) the performance framework used to measure HVDC availability;

(ii) the values that can realistically be guaranteed;

(iii) how liquidated damages are structured against those guarantees; and

(iv) a recommended strategy for American Terawatt when negotiating LDs with an equipment supplier (OEM).

2.3 Scope

- **In scope:** availability- and performance-based liquidated damages for the HVDC transmission system (converter stations and DC line), assessed under CIGRE TB 956.
- **Out of scope:** construction delay damages, defect liability, and the detailed PPA terms with the end customer (referenced only where they inform the LD basis).

2.4 Definitions and abbreviations

- **LD** — Liquidated Damages.
- **OEM** — Original Equipment Manufacturer (HVDC supplier).
- **PPA** — Power Purchase Agreement (with the end customer).
- **EA** / **EU** — Energy Availability / Unavailability (%).
- **FEU** / **SEU** — Forced / Scheduled Energy Unavailability (%).
- **FOR** — Forced Outage Rate (outages per year).
- **ODF** — Outage Derating Factor; **EOD/EOH** — Equivalent Outage Duration / Hours.

Sources: CIGRE TB 956 (2025, supersedes TB 590); American Terawatt internal review with R. Rosenqvist, weekly session 30 July 2026, and R&D team sync 29 July 2026.

3 Performance measurement framework (CIGRE TB 956)

CIGRE Technical Brochure 956 (2025), which supersedes TB 590, defines a standardised protocol for reporting the annual operational performance of HVDC systems. The reporting period is the calendar year; only events causing a loss of transfer capability are counted. These definitions are the basis on which availability guarantees — and hence liquidated damages — are written.

3.1 Outage classification

- **Scheduled outage** — planned or deferrable (e.g. preventive maintenance).
- **Forced outage** — equipment unavailable outside a scheduled state, whether by protective trip or other forced derating.
- **Forced Outage Rate (FOR)** — number of complete or partial forced outages of a converter (or pole) per unit of operating time.

3.2 Capacity and duration

- **Rated capacity** P_m — maximum continuous capacity (MW), excluding redundant equipment.
- **Outage capacity** P_o — the capacity reduction an outage would cause at P_m.
- **Outage Derating Factor** $\mathrm{ODF} = \frac{P_o}{P_m}$ — 1 for a full outage, between 0 and 1 for a partial loss.
- **Equivalent Outage Duration** $\mathrm{EOD} = \mathrm{AOD} \times \mathrm{ODF}$ — actual outage duration weighted by the derating.

3.3 Availability and unavailability

With period hours PH (8760 h/yr) and equivalent outage hours $\mathrm{EOH} = \sum \mathrm{EOD}$:

$$\mathrm{EU}\,(\%) = \frac{\mathrm{EOH}}{\mathrm{PH}} \times 100, \qquad \mathrm{EA}\,(\%) = 100 - \mathrm{EU}.$$

Unavailability is split by cause into **Forced** (FEU) and **Scheduled** (SEU) energy unavailability:

$$\mathrm{FEU} = \frac{\mathrm{EFOH}}{\mathrm{PH}} \times 100, \qquad \mathrm{SEU} = \frac{\mathrm{ESOH}}{\mathrm{PH}} \times 100.$$

FEU, SEU and FOR are the metrics against which performance guarantees are typically written.

4 Guaranteed performance values

4.1 Typical vs. guaranteed

Indicative ranges from the CIGRE biennial HVDC reliability surveys, together with values that an OEM can realistically stand behind:

Metric	Description	Typical	Guaranteed
FEU	Forced energy unavailability	0.2–1.0 %	≈0.5 %
SEU	Scheduled energy unavailability	1–3 %	≈1.0 %
FOR	Forced outage rate (pole)	4–8 / yr	2–3 / yr
	Forced outage rate (bipole)	≲0.5 / yr	≈0.1 / yr

4.2 Interpretation

- Actual figures vary with technology (LCC vs. VSC), scheme age and transmission medium: long overhead lines raise forced-outage counts, cable schemes lower them.
- SEU is driven mainly by valve-maintenance duration and can be reduced by multi-shift working during outages.
- Very aggressive bipole figures (e.g. 0.1 outages/year, or 1 in 20 years) have been *guaranteed on some projects but are commercially driven, not technically justified*; the residual exposure is effectively carried as commercial risk through the LD regime.
- VSC bipolar schemes with a dedicated metallic return are still rare, so CIGRE survey data for them is limited — guarantees carry more uncertainty.

4.3 RTEi availability-planning inputs

Additional availability-planning data provided by RTE International (RTEi) gives configuration-specific forced-outage rates and repair times:

Configuration	Outage type	FOR	MTTR
SMP	Station / pole outage	0.8 / (yr·station)	4–24 h
BIP with DMR	Pole outage	0.8 / (yr·station)	4–24 h
BIP with DMR	Bipole (full) outage	0.04–0.06 / yr	—

- **Pole/station outages are frequent but shallow.** At ≈0.8 per year and station, a pole or single-station forced outage is expected roughly once a year, but with a short repair time (MTTR 4–24 h) and, for a bipole, only a *partial* (single-pole) loss of capacity.
- **Bipole (full) outages are rare.** A forced bipole outage rate of 0.04–0.06 per year corresponds to **one event every 15–25 years** — more optimistic than the ≈0.1/yr (≈1 in 10 years) figure used elsewhere in this report, and consistent with the DMR configuration's redundancy.

- **Implication for LDs.** The high pole-outage frequency drives FOR exposure and repair logistics, while the bipole full-outage rate governs the deep-unavailability events that dominate the data-center availability case. Both should be reflected separately in the guarantee and LD structure.

Source: RTE International (RTEi) availability planning.

4.4 Monitoring period

- A monitoring period of **3 years** is typical.
- A robust buyer approach (common in North American projects) is to **extend to 4 years and count the best 3 of 4**, protecting against a single anomalous year.
- A simple mean/median-over-the-period approach is generally too risky for the buyer and should be avoided.

5 Liquidated damages mechanism

5.1 Definition and trigger

Liquidated damages are a pre-agreed sum, stated in the supply contract, payable by the OEM when a measured performance metric *exceeds* its contractually guaranteed threshold over the monitoring period. They provide commercial certainty and avoid disputes over the quantum of actual loss.

Assessment is typically **per unit of excess**:

- **FOR** — per additional forced outage above the guarantee.
- **FEU** — per increment (e.g. each additional 0.1 %) of forced energy unavailability above the guarantee.

5.2 Market ranges

Indicative LD rates for a HVDC scheme of order 1 GW:

Metric	**Penalty rate**	**Notes**
FOR (per outage)	\$0.1–1.2 M	per additional forced outage
FEU	\$75–600 k	per additional 0.1 % unavailability
LD cap	3–5 %	of contract value

5.3 Market context

- The high end of each range reflects demanding customers in a buyer's market; the low end reflects supplier-favourable conditions.
- Caps have compressed over time: 3–5 % today versus 10–20 % in the buyer's market of ∼20 years ago.
- LDs are a *sole-remedy* cap on operational performance in most contracts; the interaction with defect-liability and termination remedies must be checked case by case.

6 Availability gap and data-center context

The LD position cannot be set in isolation from the end-customer requirement. Data-center loads are specified against the Uptime Institute tier framework, and there is a structural gap between what a single HVDC link delivers and what those frameworks demand.

6.1 The tier requirement

- The industry norm for large data centers is **Tier III** (≈99.982 % availability, ≈1.6 h downtime/year).
- Tier IV (full redundancy) is generally reserved for financial institutions and is not the target here.
- Banks and project financiers require this availability analysis as a condition of financing.

6.2 The gap

- A single HVDC link **cannot** meet the Tier III requirement on its own.
- Even dual symmetrical-monopole configurations must include **tower/line failure** in the availability calculation — a factor confirmed by the DC-line development team, whose assumed tower-failure rate was significantly higher than expected, and one that is easily overlooked (it is not always captured in single-contingency studies, but most ISOs include it in their evaluation of a double circuit).

6.3 Tower failure and corridor sharing

Where redundancy is provided by two circuits, tower failure becomes a key consideration:

- **Tower spacing.** If both circuits share a structure, a single tower failure takes out both. The towers (or lines) must therefore be spaced far enough apart that a *falling tower does not affect the adjacent circuit*.
- **Shared corridor.** Spacing alone is not sufficient: as long as both circuits remain in the **same corridor / right-of-way**, a **total outage remains possible from a common-cause event** such as wildfire or another natural disaster. A wildfire, for example, can take out a parallel circuit hundreds of metres — even a mile — away through smoke/pollution flashover; a tornado through a shared corridor has historically taken out both bipoles in the same right-of-way (Manitoba experience, which led to the third bipole being routed separately).
- **Implication.** In areas of significant wildfire or severe-weather risk, true redundancy may require *separate corridors* or *undergrounding* the circuits — at a significant cable cost premium, though still potentially cheaper than local generation.

6.4 Mitigation paths

1. **Dual HVDC circuits**, each sized for full capacity (cf. the Johan Sverdrup II approach).
2. **On-site back-up generation** — effective but costly and a challenge to the scheme's green credentials.

3. **Negotiate the availability target down** with the data center — feasibility depends on the counterparty and its financing.

The chosen mitigation directly shapes the LD regime: the more the redundancy strategy relies on the HVDC supplier's performance, the more weight the LDs must carry.

6.5 Manned vs. unmanned converter station

Some operators argue that permanently manning a converter station improves availability, since staff on site can respond faster to an event. In practice this benefit is doubtful for a single project:

- **Motivation.** It is very difficult to keep qualified personnel motivated at a converter station: genuine incidents are rare and the role is largely routine maintenance, so skilled staff become bored and do not stay.
- **Skills.** Those willing to remain in such a role tend not to be the people who can provide a proper response in a real emergency; and once skilled staff leave, replacements have little of the requisite expertise — so a permanently manned station adds cost without reliably adding capability.
- **Better approach.** Retain *access* to skilled personnel with a plan to mobilise them quickly to the site, and/or an OEM service-level agreement covering maintenance, rather than permanently manning the station. Building an in-house maintenance team only becomes worthwhile across a portfolio of multiple DC projects.

7 Liquidated damages strategy

7.1 Design principles

1. **Express LDs as a percentage of OEM supply value, not absolute dollars.** OEMs assess exposure relative to project value; a percentage basis is more persuasive and scales correctly.
2. **Set LDs high enough to change behaviour.** If the penalty is too low it becomes more convenient for the OEM to pay out and deprioritise American Terawatt's project in favour of higher-penalty contracts. The purpose of the LD is to *incentivise the OEM to fix problems*.
3. **Do not expect full recovery.** LDs will not recover the true commercial loss from unavailability; they are a behavioural lever, and the residual exposure must be managed elsewhere (redundancy, storage).

7.2 Recommended approach

- **Anchor the LD basis to the PPA.** Research what the data center charges American Terawatt for unavailability under the PPA, and use a fair share of that as the LD basis — a concrete, defensible argument to the OEM.
- **Quantify with battery cost.** Frame the FEU LD around the cost of the additional battery capacity needed to compensate for forced energy-unavailability events; this gives a strong, quantifiable figure.
- **Argue for a higher cap.** Rather than accepting the standard 3–5 % cap, argue for a cap tied to the overall PPA liability that American Terawatt itself carries.

7.3 Residual exposure

Even with a well-designed LD regime, American Terawatt retains exposure to the difference between recovered LDs and actual loss. Mitigation levers include schedule and spares strategy, redundancy in the transmission configuration, and battery/on-site generation sizing.

8 Conclusions and recommendations

8.1 Conclusions

- Availability and performance guarantees — measured per CIGRE TB 956 as FEU, SEU and FOR — are the commercially decisive terms for an HVDC supply contract serving data-center load.
- Realistically guaranteeable values are FEU $\approx$0.5 %, SEU $\approx$1 % and a bipole FOR of order 1/year; more aggressive figures are commercial, not technical, commitments.
- A single HVDC link cannot meet a Tier III data-center target unaided, so the LD regime must be designed together with the redundancy and storage strategy, not after it.
- Standard LD ranges ($0.1–1.2 M per outage, $75–600 k per FEU increment, 3–5 % cap) are the starting point, not the target.

8.2 Recommendations

1. Set LDs as a percentage of OEM supply value, sized to incentivise remediation.
2. Anchor the LD basis to PPA unavailability charges and to battery replacement cost.
3. Argue for an LD cap linked to the overall PPA liability rather than the default 3–5 %.
4. Adopt a 3-year (best-3-of-4) monitoring period.

8.3 Open action items

- Research the PPA unavailability penalties charged by the data center, to use as the LD basis.
- Model the battery-replacement cost as a concrete LD justification.
- Complete the overall availability model (HVDC + batteries + on-site generation).

References

1. CIGRE Technical Brochure 956 (2025), *Protocol for reporting the operational performance of HVDC systems* (supersedes TB 590).
2. IEC 60633:2019; IEC TR 62672:2018.
3. Meisingset *et al.*, *Three-terminal hybrid HVDC interconnector for offshore wind*, CIGRE 2026, Paper B4 PS1 11874.
4. American Terawatt — internal review with R. Rosenqvist, weekly session 30 July 2026; R&D team weekly sync, 29 July 2026.